Department of Computer Engineering

Technical Interview Question & Model Answer Bank

Academic Year	**2026-2027**	**Class**	**4th year**
Subject/Expert Area	**Artificial Intelligence**		
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AI Fundamentals and Intelligent Agents

Sr. No.	Question	Model Answer	Practical Example
1	What is Artificial Intelligence?	AI is the field of computing concerned with building systems that can perceive, reason, learn, solve problems and take actions to achieve goals.	A voice assistant understands a request and performs an appropriate action.
2	How is AI different from traditional programming?	Traditional programming explicitly specifies rules, whereas many AI systems infer patterns, decisions or strategies from data, knowledge or search.	A traditional spam filter may use manually written rules; an ML spam filter learns from examples.
3	What is the relationship between AI, ML and Deep Learning?	AI is the broader field. ML is an approach within AI that learns from data, while Deep Learning uses multi-layer neural networks for learning complex representations.	Image recognition can be an AI application implemented using a deep CNN.
4	Can an AI system exist without Machine Learning?	Yes. AI also includes search, planning, rule-based reasoning, expert systems and symbolic reasoning that may not learn from data.	A chess program using Minimax and manually designed evaluation rules is an AI system without ML.
5	What is the Turing Test?	It evaluates whether a machine can produce conversational behavior that is difficult to distinguish from that of a human. It tests observable behavior rather than proving consciousness or genuine understanding.	A conversational system is evaluated through text interaction with a human evaluator.
6	What are Narrow AI, General AI and Super AI?	Narrow AI is designed for specific tasks. AGI refers to human-level general-purpose intellectual capability, while Super AI is a hypothetical concept involving capabilities beyond humans across domains.	A recommendation system is Narrow AI; AGI and Super AI remain hypothetical concepts.
7	What is an intelligent agent?	An intelligent agent perceives its environment and selects actions to achieve a goal or maximize a performance measure.	A robot vacuum senses obstacles and decides where to move next.

8	What is a rational agent?	A rational agent chooses the action expected to maximize its performance measure given its available percepts, knowledge and resources.	A navigation agent selects a route expected to minimize travel time based on available information.
9	Does a rational agent always make the correct decision?	No. Rationality depends on available information and expected outcomes. An agent may make a rational decision and still receive an unfavorable result because of uncertainty.	A navigation system chooses the fastest route, but an unexpected accident causes delay.
10	What is PEAS in AI?	PEAS stands for Performance measure, Environment, Actuators and Sensors. It is used to specify an agent's task environment.	For autonomous driving: safety/time are performance measures; roads are the environment; steering/brakes are actuators; cameras/GPS are sensors.
11	What is the difference between a simple reflex and a model-based agent?	A simple reflex agent responds only to the current percept, while a model-based agent maintains internal information about the environment.	A thermostat is simple reflex; a robot vacuum maintaining a map is model-based.
12	What is a goal-based agent?	It selects actions by considering whether they help achieve a specified goal.	A route planner searches for actions that lead from the current location to the destination.
13	What is a utility-based agent?	It selects among possible outcomes using a utility function that represents preferences or desirability.	An autonomous car may balance safety, travel time and passenger comfort.
14	What is a learning agent?	A learning agent improves its behavior from experience. It typically includes a learning component, performance element, critic and problem generator.	A game-playing agent improves its strategy after playing many games.
15	What is the difference between fully and partially observable environments?	A fully observable environment provides all information relevant to decision-making, while a partially observable environment hides some relevant information.	Chess is largely observable; driving is partially observable because obstacles can hide vehicles or pedestrians.
16	What is a deterministic versus stochastic environment?	In a deterministic environment, an action has a predictable outcome. In a stochastic environment, outcomes involve uncertainty.	A puzzle move is deterministic; weather-aware driving involves uncertainty.
17	What is a static versus dynamic environment?	A static environment does not change while the agent is deciding; a dynamic environment can change during decision-making.	A crossword puzzle is static; road traffic is dynamic.
18	What is a single-agent	A single-agent environment has one	Sudoku is primarily single-

	versus multi-agent environment?	decision-making agent, while a multi-agent environment contains multiple agents whose actions may affect each other.	agent; chess is multi-agent.

Problem Solving and Search:

Sr. No.	Question	Model Answer	Practical Example
19	What is problem formulation in AI?	It defines the initial state, actions, transition model, goal test and path cost required to solve a problem.	A route problem specifies the starting city, possible roads, destination and travel cost.
20	What is a state space?	A state space is the set of possible states reachable during problem solving.	Every possible arrangement of an 8-puzzle represents a state.
21	What is uninformed search?	It searches without domain-specific heuristic information. Examples include BFS, DFS and Uniform Cost Search.	BFS can search an unweighted maze without knowing which direction is closer to the exit.
22	What is informed search?	It uses additional knowledge, usually a heuristic, to prioritize promising states.	A route planner can use straight-line distance as a heuristic.
23	Compare BFS and DFS.	BFS explores level by level and can find a shortest path in an unweighted problem. DFS explores deeply and generally uses less memory but does not guarantee the shortest solution.	BFS is suitable for minimum-hop paths in an unweighted network.
24	When would you prefer Uniform Cost Search over BFS?	When action costs are different and the objective is to find the minimum-cost solution.	Finding the cheapest route when different roads have different travel costs.
25	What is a heuristic function?	A heuristic estimates the remaining cost from a state to the goal and helps guide search.	Straight-line distance can estimate remaining road distance.
26	What is A* search?	A* selects nodes using `f(n)=g(n)+h(n)`, where `g(n)` is the cost already incurred and `h(n)` estimates remaining cost.	GPS-style route planning can use A*-style search.
27	What makes a heuristic admissible?	It must never overestimate the actual minimum remaining cost.	Straight-line distance is an admissible heuristic for many road-distance problems.
28	Why can Greedy Best-First Search produce a poor solution?	It considers mainly the estimated distance to the goal and ignores the cost already incurred.	It may choose a geographically close city that requires a very long road route.
29	What is Hill Climbing?	It is a local search method that repeatedly moves toward a neighboring state with a	Optimizing a timetable by repeatedly making

		better evaluation value.	improvements to reduce conflicts.
30	What are local maxima, plateaus and ridges?	A local maximum is better than neighboring states but not globally best; a plateau has similar values across neighboring states; a ridge is a narrow region where direct improvement may be difficult.	Hill climbing in a complex optimization problem may stop before reaching the global optimum.
31	How does Simulated Annealing overcome local optima?	It sometimes accepts worse moves, particularly early in the search, allowing the algorithm to escape local optima.	Optimizing a large timetable or Traveling Salesman route.
32	What is Iterative Deepening Search?	It repeatedly performs depth-limited DFS with increasing depth limits, combining the low memory of DFS with the shallow-solution property of BFS.	Searching a game tree when the solution depth is unknown.
33	What is backtracking search?	It incrementally constructs a solution and abandons a partial solution as soon as it violates a constraint.	N-Queens and Sudoku are classic applications.
34	What is a Constraint Satisfaction Problem?	A CSP consists of variables, possible domains for those variables and constraints restricting valid combinations.	University timetable generation with room, teacher and time constraints.
35	Why is pruning useful in AI search?	Pruning eliminates branches that cannot produce a useful or valid solution, reducing computation.	Alpha-Beta pruning eliminates unnecessary branches from a game tree.
36	What is adversarial search?	It is search in environments where competing agents have conflicting objectives.	Chess and Tic-Tac-Toe.
37	Explain Minimax in simple terms.	Minimax chooses a move assuming that the opponent will also make the best possible move. One player maximizes utility while the opponent minimizes it.	A Tic-Tac-Toe AI evaluates possible future moves and opponent responses.
38	What is Alpha-Beta pruning?	It removes game-tree branches that cannot influence the final Minimax decision. It produces the same decision as Minimax while evaluating fewer nodes.	A chess program can search deeper by avoiding irrelevant branches.
39	What is the difference between global and local search?	Global search explores paths/states systematically, whereas local search focuses on improving a current candidate solution.	A* is global search; Hill Climbing is local search.
40	What is the exploration-exploitation trade-off?	Exploration tries unfamiliar possibilities, while exploitation uses currently known good choices. A good AI strategy balances both.	A recommendation system can occasionally recommend new items instead of always showing familiar choices.

Knowledge Representation and Reasoning:

Sr. No.	Question	Model Answer	Practical Example
41	What is knowledge representation?	It is the process of representing facts, relationships, rules and concepts so that an AI system can reason with them.	A medical expert system represents symptoms, diseases and diagnostic rules.
42	What is symbolic AI?	Symbolic AI represents knowledge explicitly using symbols, rules and logical relationships.	An expert system containing IF-THEN diagnostic rules.
43	What is the difference between symbolic and connectionist AI?	Symbolic AI relies on explicit representations and reasoning; connectionist AI learns distributed representations through neural networks.	Rule-based diagnosis versus neural-network-based image diagnosis.
44	What is propositional logic?	It represents statements that are either true or false and combines them using logical operators.	"The server is running" can be represented as a proposition.
45	Why is predicate logic more expressive than propositional logic?	Predicate logic can represent objects, properties, variables and relationships between objects.	"Every student has an ID" is naturally expressed using predicates and variables.
46	What is inference in AI?	Inference derives new knowledge from known facts and rules.	From "All humans are mortal" and "Ravi is human," infer that Ravi is mortal.
47	What is forward chaining?	It starts from known facts and repeatedly applies applicable rules to derive new facts.	A diagnostic expert system starts with observed symptoms and derives possible conditions.
48	What is backward chaining?	It starts with a goal or hypothesis and works backward to identify supporting facts and rules.	A diagnosis system starts with "Does the patient have disease X?" and checks supporting evidence.
49	What is a rule-based expert system?	It uses a knowledge base of facts/rules and an inference engine to derive conclusions.	A troubleshooting system identifies possible network failures using IF-THEN rules.
50	What is fuzzy logic?	Fuzzy logic represents degrees of truth rather than only true/false values, making it useful for vague concepts.	An air conditioner can interpret temperature as "slightly hot" or "very hot."
51	How is fuzzy logic different from	Boolean logic uses discrete true/false values, whereas fuzzy logic represents	"Temperature is high" may be represented with a

	Boolean logic?	degrees of membership or truth.	membership value such as 0.7 in a fuzzy controller.
52	When would you use probabilistic reasoning instead of deterministic reasoning?	When available information is uncertain, incomplete or noisy, probabilities can represent degrees of belief.	A medical system may calculate the probability of a disease from symptoms.
53	What is a Bayesian Network?	It is a probabilistic graphical model representing variables and conditional dependencies using a directed acyclic graph.	A diagnostic network can represent relationships among symptoms and diseases.
54	Why is uncertainty important in AI?	Real-world environments often contain incomplete, noisy or ambiguous information, so AI systems need methods for reasoning despite uncertainty.	A self-driving system may have uncertain information about another vehicle's intention.

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Machine Learning

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55	What is Machine Learning?	ML enables systems to learn patterns from data and use those patterns to make predictions or decisions without explicitly programming every rule.	Learning to classify emails as spam or legitimate.
56	What is supervised learning?	It learns from labelled input-output examples to predict outputs for new	Predicting whether a loan application will default.

		inputs.	
57	What is unsupervised learning?	It discovers structure or patterns in data without target labels.	Grouping customers based on purchasing behavior.
58	What is reinforcement learning?	An agent learns through interaction with an environment using rewards and penalties to improve its policy.	A game-playing agent learns which actions lead to winning.
59	What is semi-supervised learning?	It uses a small amount of labelled data together with a larger amount of unlabelled data.	Classifying thousands of images when only a small subset has been manually labelled.
60	What is the difference between classification and regression?	Classification predicts categories; regression predicts continuous numerical values.	Disease/no-disease is classification; predicting blood pressure is regression.
61	What is clustering?	Clustering groups data points based on similarity without predefined labels.	Segmenting customers into behavioral groups.
62	What is overfitting?	The model learns training-specific patterns or noise and performs poorly on unseen data.	99% training accuracy but 72% test accuracy.
63	What is underfitting?	The model is too simple to capture the important patterns and performs poorly even on training data.	A linear model failing to capture a strongly nonlinear relationship.
64	Explain the bias-variance trade-off.	High bias generally indicates an overly simple model, while high variance indicates excessive sensitivity to training data. Good model selection balances both.	Comparing a very simple linear model with an overly complex decision tree.
65	Why do we use training, validation and test datasets?	Training learns the model, validation helps select/tune it, and the test set provides a final estimate on unseen data.	Comparing different hyperparameter settings using validation data and evaluating the final model on a separate test set.
66	What is data leakage?	Data leakage occurs when information unavailable at prediction time unintentionally enters the training process, producing unrealistically good results.	Using a future medical test result as an input when predicting an earlier diagnosis.
67	What is cross-validation?	It repeatedly divides the training data into training and validation portions to estimate model performance more reliably.	Five-fold cross-validation for comparing classification models.
68	What is feature engineering?	It involves creating or transforming input features to make useful	Extracting "number of purchases in the last 30 days"

		patterns easier for a model to learn.	from transaction history.
69	Why is feature scaling needed?	Some algorithms are sensitive to feature magnitudes. Scaling prevents large-valued features from dominating the learning process.	KNN using age and annual income benefits from appropriate scaling.
70	What is normalization versus standardization?	Normalization commonly maps values to a specified range, while standardization transforms values using mean and standard deviation.	Scaling numerical features before KNN or neural-network training.
71	What is regularization?	Regularization adds a penalty or constraint to discourage overly complex models and reduce overfitting.	L1 or L2 regularization in a regression model.
72	What is the difference between L1 and L2 regularization?	L1 penalizes the absolute values of weights and can produce sparse models; L2 penalizes squared weights and generally shrinks weights smoothly.	L1 can help select a subset of useful features.
73	What is a confusion matrix?	It summarizes classification results using true positives, true negatives, false positives and false negatives.	Evaluating a fraud-detection classifier.
74	When is accuracy a poor metric?	Accuracy can be misleading for imbalanced datasets. Other metrics such as precision, recall, F1 or PR-AUC may be more informative.	A fraud dataset with 99% legitimate transactions.
75	What is precision?	Precision is the proportion of predicted positives that are actually positive: `TP/(TP+FP)`.	Of transactions flagged as fraud, how many are truly fraudulent?
76	What is recall?	Recall is the proportion of actual positives that the model correctly identifies: `TP/(TP+FN)`.	Of all fraudulent transactions, how many did the system detect?
77	What is F1-score?	F1 is the harmonic mean of precision and recall and is useful when both types of classification errors matter.	Evaluating an imbalanced spam classifier.
78	How would you handle an imbalanced dataset?	Possible approaches include suitable metrics, resampling, class weights, threshold adjustment and algorithms that handle imbalance effectively.	Fraud detection with very few fraudulent transactions.
79	What is a Decision Tree?	It recursively splits data using feature-based conditions to make predictions.	Predicting whether a customer will purchase a product.
80	Why can a Decision Tree	A tree can become excessively deep	A tree with a separate branch

	overfit?	and learn noise or very specific training cases. Pruning, depth limits and other controls can reduce this.	for nearly every training record.
81	What is Random Forest?	Random Forest combines many decision trees trained using randomized data/features and aggregates their predictions, generally improving robustness over a single tree.	Customer churn prediction.
82	What is boosting?	Boosting builds models sequentially, with later models focusing on errors or residuals from earlier models.	Gradient Boosting for tabular classification.
83	What is K-Nearest Neighbors?	KNN predicts based on the labels or values of nearby training examples according to a distance measure.	Classifying a new customer based on similar existing customers.
84	What is the basic idea of SVM?	SVM seeks a decision boundary that separates classes while maximizing the margin between them. Kernels can enable nonlinear boundaries.	Binary classification of text or structured data.
85	What is Naïve Bayes?	It is a probabilistic classifier based on Bayes' theorem with a simplifying conditional-independence assumption.	Spam classification based on word occurrence patterns.

Neural Networks and Deep Learning

Sr. No.	Question	Model Answer	Practical Example
86	What is an artificial neuron?	It computes a weighted combination of inputs, adds a bias and passes the result through an activation function.	A neuron can combine image features to help classify an object.
87	Why are activation functions required?	They introduce non-linearity, allowing neural networks to model complex relationships.	ReLU is commonly used in hidden layers of deep networks.
88	What is an epoch?	One epoch means the model has processed the entire training dataset once.	Training a neural network for 20 epochs means the training data is processed 20 times, subject to the training setup.
89	What is batch size?	Batch size is the number of training samples processed before an optimizer update.	A batch size of 32 means weights are typically updated after processing 32 samples.

90	What is learning rate?	It controls the magnitude of parameter updates during optimization.	An excessively high learning rate can cause unstable training.
91	What is gradient descent?	It iteratively updates model parameters in the direction that reduces the loss function.	Training a neural network by minimizing prediction error.
92	What is backpropagation?	It efficiently computes gradients of the loss with respect to network parameters using the chain rule, enabling weight updates.	Training a neural network to recognize handwritten digits.
93	What is the vanishing-gradient problem?	Gradients can become extremely small as they propagate through deep networks, slowing or preventing learning in earlier layers.	Older deep RNNs could struggle to learn long-range dependencies.
94	What is a CNN and why is it useful for images?	CNNs use convolutional operations to learn local spatial patterns and hierarchical visual features efficiently.	Image classification and object detection.
95	What is pooling in CNNs?	Pooling reduces spatial dimensions and can provide some translation robustness while reducing computation.	Max pooling can reduce the dimensions of feature maps after convolution.
96	What is an RNN?	An RNN processes sequential information while maintaining a hidden state representing information from previous steps.	Processing a sequence of words or time-series observations.
97	Why were LSTM and GRU architectures introduced?	They address some limitations of basic RNNs in learning long-term dependencies using gating mechanisms.	Predicting patterns in long sequences.
98	What is transfer learning?	A model trained on one task/domain is reused as a starting point for another related task.	Fine-tuning a pretrained image model for defect detection.
99	Why is transfer learning useful for fresh AI projects?	It can reduce training time and data requirements because the model already contains useful learned representations.	Fine-tuning a pretrained vision model with a small college project dataset.
100	What is the difference between training from scratch and fine-tuning?	Training from scratch learns parameters from random initialization, while fine-tuning starts with an already trained model and adapts it to a new task.	Fine-tuning a pretrained language or vision model instead of building one from zero.

NLP, Transformers and Generative AI

Sr. No.	Question	Model Answer	Practical Example
101	What is Natural Language Processing?	NLP is the AI field concerned with processing, understanding and generating human language.	Chatbots, translation and sentiment analysis.
102	What is tokenization?	Tokenization divides text into smaller units such as words, subwords or tokens that a model can process.	A sentence is converted into a sequence of tokens before entering an LLM.
103	What is the difference between stemming and lemmatization?	Stemming uses simpler rules to reduce words, while lemmatization attempts to produce the linguistically correct base form.	"running" may be reduced to "run" through lemmatization.
104	What are word embeddings?	They represent words or tokens as dense numerical vectors that capture semantic relationships.	Words with related meanings can have similar vector representations.
105	What is the limitation of traditional static word embeddings?	A static embedding assigns essentially the same representation to a word regardless of context, so it can struggle with words having multiple meanings.	"Bank" has different meanings in "river bank" and "bank account."
106	What is attention in deep learning?	Attention allows a model to assign different importance to different input elements when constructing a representation or generating an output.	In a sentence, the model can focus on relevant words when interpreting a pronoun.
107	What is a Transformer?	A Transformer is a neural architecture built around attention mechanisms that processes sequence elements efficiently and captures relationships between them.	Modern language models use Transformer-based architectures.
108	Why are Transformers important compared with traditional RNNs?	They allow substantial parallel processing during training and can model long-range relationships more effectively than basic recurrent architectures.	Large language models can process long textual contexts using Transformer architectures.
109	What is self-attention?	Self-attention allows each token to consider other tokens in the same input sequence when creating its representation.	The meaning of a word can be influenced by surrounding words.
110	What are Query, Key and Value in attention?	Query represents what a token is looking for, Key represents what	A token can attend more strongly to another token

		each token offers for matching, and Value contains information aggregated according to those attention scores.	whose key matches its query.
111	What is an LLM?	A Large Language Model is a large neural language model trained on extensive text data to model and generate language.	An LLM can answer questions, summarize text or generate code.
112	How does an LLM generate text?	It generally predicts the probability of the next token based on preceding context and repeatedly generates tokens to form an output.	Given "The capital of India is", the model predicts likely next tokens.
113	What is Generative AI?	Generative AI creates new content such as text, images, audio, video or code based on learned patterns.	Generating a product description from a few specifications.
114	What is hallucination in an LLM?	Hallucination occurs when a model produces plausible-sounding information that is unsupported or factually incorrect.	An LLM inventing a research paper citation that does not exist.
115	Why do LLMs hallucinate?	They generate likely sequences based on learned patterns rather than inherently verifying every factual statement against a trusted source.	A model may confidently generate an incorrect historical detail.
116	What is prompt engineering?	It involves designing instructions and context to obtain more useful, reliable and appropriately structured model outputs.	Asking an LLM to return a response in a specified JSON structure.
117	What is zero-shot prompting?	The model is asked to perform a task without providing examples.	"Classify this review as positive or negative."
118	What is few-shot prompting?	A prompt provides a small number of examples demonstrating the desired task or output format.	Giving three examples of sentiment classification before asking for a new classification.
119	What is RAG?	Retrieval-Augmented Generation retrieves relevant external information and supplies it to an LLM as context before generating a response.	A college chatbot retrieves information from university regulations before answering students.
120	Why can RAG be preferable to relying only on an LLM's internal knowledge?	RAG can provide current or domain-specific information without changing the model's weights and can ground responses in retrieved documents.	A company chatbot can answer questions using the latest internal HR policies.

121	What is an embedding in a RAG system?	An embedding converts text or another object into a numerical vector representing semantic information, enabling similarity-based retrieval.	A user query can be compared with document embeddings to find relevant passages.
122	What is a vector database?	It stores and retrieves vector representations efficiently, usually using similarity search.	A RAG application searches a collection of company documents using embedding similarity.
123	RAG versus fine-tuning: what is the basic difference?	RAG supplies external information at inference time, while fine-tuning changes model parameters through additional training.	Frequently changing company policies are often suitable for retrieval; specialized behavior may motivate fine-tuning.
124	How would you reduce hallucinations in a RAG application?	Improve retrieval quality, use reliable source documents, constrain the prompt, require answers to be grounded in retrieved content, and evaluate responses systematically.	A legal-document chatbot can answer only from retrieved approved documents.

Reinforcement Learning

Sr. No.	Question	Model Answer	Practical Example
125	What are the main components of Reinforcement Learning?	Agent, environment, state, action, reward and policy are core components; value functions and environment models may also be used.	A game-playing agent observes the game state, selects a move and receives a reward.
126	What is a policy in reinforcement learning?	A policy specifies how an agent chooses actions based on the current state.	A robot policy determines whether it should move forward, turn or stop.
127	What is the exploration-exploitation problem in RL?	The agent must balance trying unknown actions with using actions already believed to be good.	A recommendation system may explore a new product while continuing to recommend popular products.
128	What is Q-Learning?	Q-Learning is a model-free RL algorithm that learns the expected long-term value of taking an action in a state.	Learning optimal movements through a grid-world.
129	What is the difference between on-policy and off-policy learning?	On-policy methods learn about the policy being followed, whereas off-policy methods can learn an optimal policy from experiences generated by another behavior policy.	SARSA is on-policy; Q-Learning is off-policy.

130	What is a reward function?	It defines the immediate feedback the environment gives an agent after an action.	A game may give +1 for winning and -1 for losing.
131	Why is reward design important?	A poorly designed reward can cause the agent to optimize an unintended behavior even if it technically maximizes the reward.	A robot rewarded only for speed might ignore safety.
132	What is the discount factor?	It determines how strongly future rewards influence the current decision compared with immediate rewards.	A high discount factor encourages consideration of long-term consequences.

Computer Vision and Modern AI

Sr. No.	Question	Model Answer	Practical Example
133	What is Computer Vision?	It is an AI field concerned with extracting useful information and understanding from images and videos.	Detecting vehicles in traffic-camera footage.
134	What is image classification?	It assigns one or more labels to an entire image.	Classifying an image as cat, dog or bird.
135	What is object detection?	It identifies objects and their locations, commonly using bounding boxes.	Detecting pedestrians and vehicles in a road image.
136	What is image segmentation?	It assigns labels to individual pixels or regions of an image.	Separating tumor tissue from healthy tissue in a medical image.
137	Why are CNNs widely used in computer vision?	Their convolutional structure efficiently learns local and hierarchical spatial features.	Recognizing handwritten digits or detecting objects.
138	What is data augmentation in computer vision?	It creates modified training examples through transformations such as rotation, cropping or flipping to improve generalization.	Slightly rotating training images of handwritten digits.
139	What is a GAN?	A GAN contains a generator that creates synthetic samples and a discriminator that tries to distinguish generated samples from real ones.	Generating synthetic face images.
140	What is a diffusion model?	It learns to generate data by learning to reverse a gradual noise-adding process.	Modern image-generation systems can generate images from text prompts using diffusion-based methods.

Responsible, Explainable and Production AI:

Sr. No.	Question	Model Answer	Practical Example
141	What is Explainable AI?	XAI includes techniques that help humans understand why an AI system produced a particular prediction or decision.	Explaining why a credit-risk model assigned a high-risk prediction.
142	Why is explainability	It can support trust, debugging,	A bank may need to explain

	important?	accountability, regulatory requirements and detection of inappropriate model behavior.	factors influencing a lending decision.
143	What is bias in AI?	AI bias is systematic behavior that produces unfair or skewed outcomes, potentially due to data, labels, features, model design or deployment conditions.	A hiring model trained on unrepresentative historical data may disadvantage some applicant groups.
144	How can bias enter an AI system?	Bias can enter through data collection, sampling, historical labels, feature selection, annotation, model design or deployment.	Historical hiring decisions can influence training labels.
145	What is model drift?	Model or concept drift occurs when relationships between inputs and outcomes change over time, reducing model performance.	Customer behavior changes after a new product launch.
146	What is data drift?	Data drift refers to changes in the distribution of input data over time.	The age distribution of users changes substantially after a product expands to a new market.
147	What is MLOps?	MLOps combines ML development with operational practices for versioning, deployment, monitoring, reproducibility and maintenance.	Automatically retraining a fraud model when monitored performance deteriorates.
148	Why should an AI model be monitored after deployment?	Real-world data, user behavior and operating conditions can change, so model performance may degrade even if the model worked well during testing.	A recommendation model may become less effective as customer preferences change.
149	Your AI model has excellent test accuracy but poor production performance. What would you investigate?	I would check data drift, preprocessing differences, training-serving mismatch, leakage, changes in the target relationship, production data quality and infrastructure issues.	A fraud model trained on old transaction patterns may fail after new fraud techniques appear.
150	An LLM gives incorrect answers from a company's internal documents. How would you investigate?	I would inspect document quality, chunking, embeddings, retrieval relevance, context limits, prompt design and generation behavior. I would evaluate retrieval separately from answer generation.	An HR chatbot retrieves an outdated policy instead of the current approved policy.